

**A HYBRID MACHINE LEARNING-BASED SMART HOME IOT
SECURITY SYSTEM FOR THREAT DETECTION, CLASSIFICATION,
AND ALERT MECHANISMS**

**A PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF**

**BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING**

BY

GADIVENUKA ABHISHEK GOUD	22071A0518
GOURISHETTY SAI SRUJITH	22071A0521
VADAPARTHY PAVAN KUMAR	22071A0559
YAMSANI SAKETH	22071A0564

UNDER THE GUIDANCE OF

Mr. P PRAVEEN

ASSISTANT PROFESSOR

DEPT. OF COMPUTER SCIENCE & ENGINEERING



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

VALLURUPALLI NAGESWARA RAO

VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY

Vignana Jyothi Nagar, Bachupally, Hyderabad 500118, Telangana India

March 2026



VNRVJIET

VALLURUPALLI NAGESWARA RAO
VIGNANA JYOTHI INSTITUTE OF
ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**

CERTIFICATE

This is to certify that the Major Project titled “**A HYBRID MACHINE LEARNING-BASED SMART HOME IOT SECURITY SYSTEM FOR THREAT DETECTION, CLASSIFICATION, AND ALERT MECHANISMS**” is a record of bona fide work submitted by:

- **GADIVENUKA ABHISHEK GOUD** 22071A0518
- **GOURISHETTY SAI SRUJITH** 22071A0521
- **VADAPARTHY PAVAN KUMAR** 22071A0559
- **YAMSANI SAKETH** 22071A0564

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech.)** in **COMPUTER SCIENCE AND ENGINEERING** of **Jawaharlal Nehru Technological University Hyderabad (JNTUH)**, under **VNRVJIET B.Tech. R22 Academic Regulations**. The work has been carried out under our guidance and supervision at **Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering & Technology, Hyderabad**.

The report is original, free from plagiarism, and complies with the Institute guidelines, including those related to academic ethics and AI-generated content.

Signature of Supervisor

Mr. P Praveen

Assistant Professor

Computer Science & Engineering

VNR VJIET, Hyderabad

Signature of HOD

Dr. V Baby

Associate Professor & HOD

Computer Science & Engineering

VNR VJIET, Hyderabad



VNRVJET

**VALLURUPALLI NAGESWARA RAO
VIGNANA JYOTHI INSTITUTE OF
ENGINEERING & TECHNOLOGY**

AUTONOMOUS INSTITUTION

**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**

PLAGIARISM CERTIFICATE

This is to certify that the Major Project report titled "**A HYBRID MACHINE LEARNING-BASED SMART HOME IOT SECURITY SYSTEM FOR THREAT DETECTION, CLASSIFICATION, AND ALERT MECHANISMS**" submitted by **GADIVENUKA ABHISHEK GOUD (22071A0518), GOURISHETTY SAI SRUJITH (22071A0521), VADAPARTHY PAVAN KUMAR (22071A0559), YAMSANI SAKETH (22071A0564)** has been checked in "**TURNITIN**" software for AI plagiarism and similarity index is within the permissible limit as per the Institute norms.

AI Plagiarism: <20%

Similarity Index: 4%

Signature of Supervisor

Mr. P PRAVEEN

Assistant Professor,

Department of Computer Science and Engineering

VNR VJET, Hyderabad

APPROVAL CERTIFICATE

Major Project evaluation for the project titled “A HYBRID MACHINE LEARNING-BASED SMART HOME IOT SECURITY SYSTEM FOR THREAT DETECTION, CLASSIFICATION, AND ALERT MECHANISMS”, being submitted by **Mr. GADIVENUKA ABHISHEK GOUD**, Regd. No. **22071A0518**, **Mr. GOURISHETTY SAI SRUJITH**, Regd. No. **22071A0521**, **Mr. VADAPARTHY PAVAN KUMAR**, Regd. No. **22071A0559**, **Mr. YAMSANI SAKETH**, Regd. No. **22071A0564**, has been conducted on _____ and the work is approved for the award of the degree of **BACHELOR OF TECHNOLOGY (B.Tech.) in COMPUTER SCIENCE AND ENGINEERING**, under VNRVJIET B.Tech. R22 Academic Regulations.

INTERNAL EXAMINER

EXTERNAL EXAMINER



VNRVJIET

**VALLURUPALLI NAGESWARA RAO
VIGNANA JYOTHI INSTITUTE OF
ENGINEERING & TECHNOLOGY**

AUTONOMOUS INSTITUTION

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

We, the undersigned, declare that the Major Project titled **“A HYBRID MACHINE LEARNING-BASED SMART HOME IOT SECURITY SYSTEM FOR THREAT DETECTION, CLASSIFICATION, AND ALERT MECHANISMS”** has been carried out by us in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech.) in COMPUTER SCIENCE AND ENGINEERING** of **Jawaharlal Nehru Technological University Hyderabad (JNTUH)**, under **VNRVJIET B.Tech. R22 Academic Regulations**, at **Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering & Technology, Hyderabad**.

We further declare that this work is original, has not been submitted previously for any degree or similar title, and complies with the Institute's plagiarism and academic ethics guidelines, including AI-generated content. The contributions of all team members are clearly indicated in the report.

PLACE: HYDERABAD

DATE: _____

Signature of the Student(s)

GADIVENUKA ABHISHEK GOUD

22071A0518

Abhishek

GOURISHETYY SAI SRUJITH

22071A0521

G. Srujith

VADAPARTHY PAVAN KUMAR

22071A0559

V. Pavan

YAMSANI SAKETH

22071A0564

Y. Saketh